made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselv
move forward in partnership with Indigenous communities in a spirit of Reconciliation and collaboration.

# Symmetric graph matrices and spectral clustering

## MATH 2740 – Mathematics of Data Science – Unit U10

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# Outline

# From directed rankings to undirected communities

In Units 08 and 09, we studied directed graphs, non-negative matrices and the Perron-Frobenius theorem

There, the dominant eigenvector $\boldsymbol{v}_1 > \mathbf{0}$ associated with $\rho(A)$ answered a *ranking* question: which vertices are most central or visited most often?

In this unit, we turn to **undirected graphs** and **similarity networks**, where the associated matrices are *real and symmetric*

The main question shifts from ranking individual vertices to **clustering**: how do we partition a graph (or a dataset) into cohesive communities with minimal connections between them?

# Why real symmetric matrices change everything

When a matrix $M \in \mathcal{M}_n(\mathbb{R})$ is symmetric ($M^T = M$), the spectral theorem gives us three powerful tools that general directed graphs lack:

1. Every eigenvalue $\lambda_1, \ldots, \lambda_n$ of $M$ is **real**

2. Eigenvectors from distinct eigenspaces are **mutually orthogonal**, and $\mathbb{R}^n$ has an **orthonormal basis of eigenvectors** $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$

3. Eigenvalues and eigenvectors are characterised variationally through the **quadratic form** $\boldsymbol{x}^T M \boldsymbol{x}$ and the **Rayleigh quotient** $\dfrac{\boldsymbol{x}^T M \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}}$

# Roadmap of Unit 10

1. **Undirected graphs and similarity graphs:** turning point clouds $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$ into symmetric adjacency and weight matrices

2. **The combinatorial graph Laplacian $L = D - A$:** degree matrix, incidence factorisation $L = B^T B$ and the quadratic form $\boldsymbol{x}^T L \boldsymbol{x} = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2$

3. **Spectral properties of $L$:** positive semi-definiteness, $\lambda_1 = 0$ with eigenvector $\mathbb{1}$ and why the multiplicity of $\lambda = 0$ counts connected components

4. **Graph cuts, the Fiedler vector and spectral clustering:** expressing $\mathrm{cut}(S, \bar{S}) = \frac{1}{4} \boldsymbol{x}^T L \boldsymbol{x}$, spectral relaxation and partitioning with the Fiedler vector $\boldsymbol{v}_2$

# Undirected simple graphs

Let $G = (V, E)$ be an undirected simple graph with vertex set $V = \{1, 2, \ldots, n\}$ and edge set $E$ consisting of unordered pairs $\{i, j\}$ of distinct vertices

- Undirected means $\{i, j\} = \{j, i\}$: if $i$ is connected to $j$, then $j$ is connected to $i$
- Simple means there are no self-loops ($\{i, i\} \notin E$) and at most one edge between any pair of vertices

**Definition 155 (Adjacency matrix of an undirected graph)**

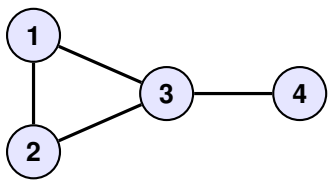
*The **adjacency matrix** of an undirected simple graph $G = (V, E)$ on $n$ vertices is the matrix $A = [a_{ij}] \in \mathcal{M}_n(\{0, 1\})$ defined by*

$$a_{ij} = \begin{cases} 1 & \text{if } \{i, j\} \in E \\ 0 & \text{otherwise} \end{cases}$$

# Symmetry of the adjacency matrix

Because $\{i,j\} \in E \iff \{j,i\} \in E$ and $\{i,i\} \notin E$, the adjacency matrix $A$ of an undirected simple graph always satisfies

$$A^T = A \quad (a_{ij} = a_{ji} \ \ \forall i,j) \qquad \text{and} \qquad a_{ii} = 0 \quad (\forall i = 1, \ldots, n)$$




$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = A^T$$

Notice that the trace of $A$ is $\text{tr}(A) = \sum_{i=1}^{n} a_{ii} = 0$, so the $n$ real eigenvalues of $A$ always sum to 0 (some are positive and some are negative)

# Weighted similarity graphs

In data science, edges often carry a nonnegative weight measuring the **similarity** (or affinity) between vertices $i$ and $j$

**Definition 156 (Weighted similarity graph and weight matrix)**

A **weighted undirected graph** $G = (V, E, W)$ has a symmetric weight matrix $W = [w_{ij}] \in \mathcal{M}_n(\mathbb{R}_{\geq 0})$ satisfying

$$w_{ij} = w_{ji} \geq 0 \quad (\forall i, j = 1, \ldots, n) \qquad \text{and} \qquad w_{ii} = 0 \quad (\forall i = 1, \ldots, n)$$

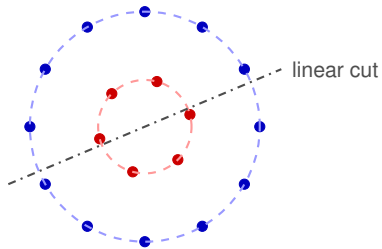
Here $w_{ij} > 0$ if $\{i, j\} \in E$ and $w_{ij} = 0$ if $\{i, j\} \notin E$

- Large $w_{ij}$ means vertices $i$ and $j$ are **strongly connected** (highly similar)
- $w_{ij} = 0$ (or $w_{ij} \approx 0$) means vertices $i$ and $j$ are **unconnected** (dissimilar)
- An unweighted graph is simply the special case $W = A \in \mathcal{M}_n(\{0, 1\})$

# Why turn a dataset into a similarity graph?

Suppose we are given $n$ data points $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$ and want to partition them into clusters

- Centroid-based methods ($k$-means) and linear projections (PCA) rely on **global Euclidean distances** $\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2$ and linear separating hyperplanes
- They work well when clusters are compact, convex clouds
- They fail completely when clusters have **non-convex geometry** (such as concentric rings, spirals or interlocking crescents), where points in the same cluster are far apart in $\mathbb{R}^d$ but connected through a chain of nearby neighbours




Representing the data as a **local similarity graph** turns non-convex geometric clustering into a **graph connectivity** problem

# Constructing similarity graphs from data points

Given data points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$ with pairwise distances $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2$, there are three standard ways to build a symmetric similarity graph:

1. **$\varepsilon$-neighbourhood graph**: Fix a threshold $\varepsilon > 0$ and connect $i \neq j$ with an unweighted edge ($a_{ij} = 1$) if and only if

$$\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 \leq \varepsilon$$

2. ***k*-nearest neighbour (*k*-NN) graph**: Connect $i$ and $j$ if $\boldsymbol{x}_i$ is among the $k$ closest points to $\boldsymbol{x}_j$ *or* $\boldsymbol{x}_j$ is among the $k$ closest points to $\boldsymbol{x}_i$ (symmetrised so $A^T = A$)
3. **Gaussian (RBF) similarity graph**: Connect all pairs $i \neq j$ with weight

$$w_{ij} = \exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2}{2\sigma^2}\right), \qquad w_{ii} = 0$$

where $\sigma > 0$ is the bandwidth (scale) parameter

# Example: from 4 points in $\mathbb{R}^2$ to a similarity graph

Consider $n = 4$ points in $\mathbb{R}^2$: two nearby points $\boldsymbol{x}_1 = (0,0)^T, \boldsymbol{x}_2 = (0,1)^T$ on the left and two nearby points $\boldsymbol{x}_3 = (3,0)^T, \boldsymbol{x}_4 = (3,1)^T$ on the right

Pairwise Euclidean distances $\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2$:

$$\Delta = \begin{pmatrix} 0 & 1 & 3 & \sqrt{10} \\ 1 & 0 & \sqrt{10} & 3 \\ 3 & \sqrt{10} & 0 & 1 \\ \sqrt{10} & 3 & 1 & 0 \end{pmatrix}$$

With $\varepsilon = 1.5$, the **$\varepsilon$-neighbourhood graph** has adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

With $\sigma = 1$, the **Gaussian similarity matrix** $w_{ij} = e^{-\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2/2}$ $(i \neq j)$ is

$$W \approx \begin{pmatrix} 0 & 0.607 & 0.011 & 0.007 \\ 0.607 & 0 & 0.007 & 0.011 \\ 0.011 & 0.007 & 0 & 0.607 \\ 0.007 & 0.011 & 0.607 & 0 \end{pmatrix}$$

- Within-cluster weights: $e^{-1/2} \approx 0.607$
- Between-cluster weights: $e^{-9/2} \approx 0.011$ and $e^{-10/2} \approx 0.007$

# Role of the scale parameter $\varepsilon$, $k$ or $\sigma$

In all three constructions, the scale parameter controls which pairs are treated as local neighbours:

**Scale too small**



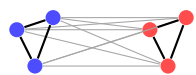
True clusters shatter into many isolated pieces

**Well-chosen scale**



Dense within clusters, sparse or zero between clusters

**Scale too large**



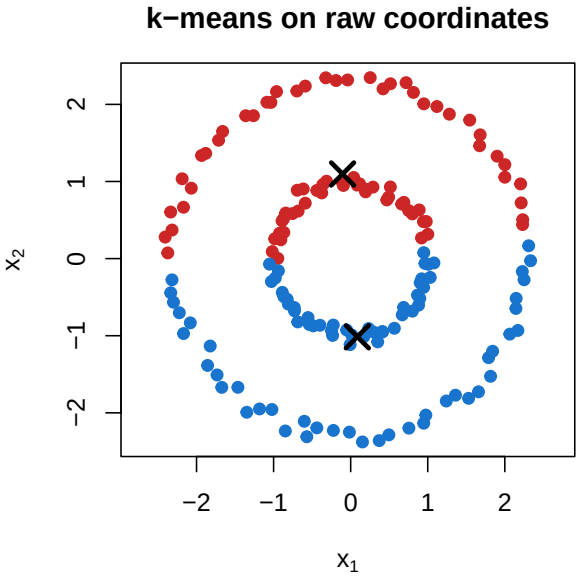
Shortcut edges merge distinct clusters together

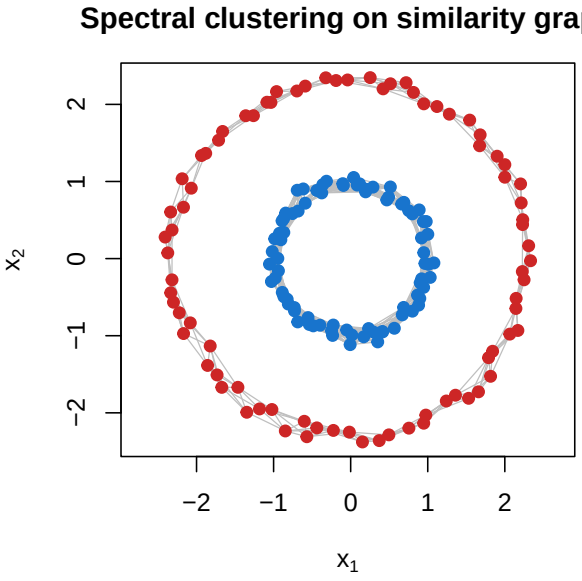
In what follows, we write $A = [a_{ij}]$ for the adjacency or similarity matrix (whether $a_{ij} \in \{0, 1\}$ or $a_{ij} = w_{ij} \geq 0$): all formulas hold identically in both cases

# R illustration: two concentric rings

Let us generate $n = 160$ points on two concentric circles in $\mathbb{R}^2$ and compare $k$-means in $\mathbb{R}^2$ against an $\varepsilon$-neighbourhood graph

```
set.seed(2740)
n1 = 80; n2 = 80
theta1 = seq(0, 2 * pi, length.out = n1 + 1)[-(n1 + 1)]
theta2 = seq(0, 2 * pi, length.out = n2 + 1)[-(n2 + 1)]
X = rbind(
  cbind(1.0 * cos(theta1), 1.0 * sin(theta1)) + rnorm(2 * n1, sd = 0.06),
  cbind(2.3 * cos(theta2), 2.3 * sin(theta2)) + rnorm(2 * n2, sd = 0.06)
)
# epsilon-neighbourhood adjacency matrix (epsilon = 0.65)
dist_mat = as.matrix(dist(X))
A_rings = (dist_mat <= 0.65 & dist_mat > 0) * 1
```


k−means on raw coordinates
x2
x1
2
1
0
−1
−2
−2
−1
0
1
2


Spectral clustering on similarity graph
x2
x1
2
1
0
−1
−2
−2
−1
0
1
2

# Vertex degrees and the degree matrix

**Definition 157 (Vertex degree and degree matrix)**

*Let $G = (V, E)$ be an undirected graph on $n$ vertices with symmetric adjacency (or weight) matrix $A = [a_{ij}] \in \mathcal{M}_n(\mathbb{R}_{\geq 0})$. The* ***degree*** *of vertex $i$ is the row sum*

$$d_i = \sum_{j=1}^{n} a_{ij} = (A\mathbb{1})_i \qquad (i = 1, \ldots, n)$$

*where $\mathbb{1} = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$. The* ***degree matrix*** *of $G$ is the diagonal matrix*

$$D = \operatorname{diag}(d_1, d_2, \ldots, d_n) = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}$$

# The combinatorial graph Laplacian

**Definition 158 (Combinatorial graph Laplacian)**

*Let $G = (V, E)$ be an undirected graph on $n$ vertices with degree matrix $D$ and adjacency (or weight) matrix $A$. The* ***combinatorial graph Laplacian*** *(or simply* ***Laplacian matrix****) of $G$ is the $n \times n$ matrix*

$$L = D - A$$

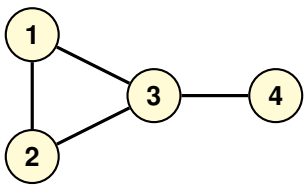
*Entry-by-entry, $L = [\ell_{ij}]$ is given by*

$$\ell_{ij} = \begin{cases} d_i & \text{if } i = j \\ -a_{ij} & \text{if } i \neq j \end{cases}$$

For a simple unweighted graph, $\ell_{ii} = d_i$, $\ell_{ij} = -1$ if $\{i, j\} \in E$ and $\ell_{ij} = 0$ otherwise

# Example 1: Laplacian of a 4-vertex paw graph

Consider the 4-vertex graph $G = (V, E)$ below (a triangle $\{1, 2, 3\}$ with a pendant edge $\{3, 4\}$):




Degrees: $d_1 = 2$, $d_2 = 2$, $d_3 = 3$, $d_4 = 1$

$$D = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

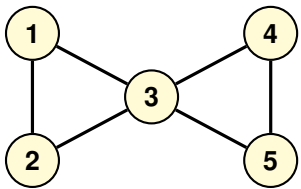
$$L = D - A = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Quick sanity check on $L$:

- $L$ is symmetric ($L^T = L$), with positive degrees on the diagonal and $-1$ on edges
- Every row and every column of $L$ sums to 0: e.g. row 3 has $(-1) + (-1) + 3 + (-1) = 0$

# Example 2: Laplacian of a 5-vertex bowtie graph

Consider the 5-vertex bowtie graph formed by two triangles $\{1, 2, 3\}$ and $\{3, 4, 5\}$ sharing central vertex 3:




Degrees: $d_1 = d_2 = d_4 = d_5 = 2$, $d_3 = 4$

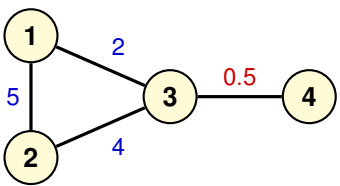
$$L = D - A = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ -1 & -1 & 4 & -1 & -1 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{pmatrix}$$

Notice how the structure of $L$ reflects the graph topology:

- The upper-left $2 \times 2$ block $\{1, 2\}$ and lower-right $2 \times 2$ block $\{4, 5\}$ have no direct edges between them (0 in the off-diagonal blocks)
- All paths between $\{1, 2\}$ and $\{4, 5\}$ pass through vertex 3 (row and column 3)

# Example 3: Laplacian of a weighted similarity graph

For a weighted graph $G = (V, E, W)$, the exact same recipe $L = D - W$ applies using weighted degrees $d_i = \sum_{j=1}^{n} w_{ij}$:




Weighted degrees:
$d_1 = 5 + 2 = 7,\ d_2 = 5 + 4 = 9$
$d_3 = 2 + 4 + 0.5 = 6.5,\ d_4 = 0.5$

$$W = \begin{pmatrix} 0 & 5 & 2 & 0 \\ 5 & 0 & 4 & 0 \\ 2 & 4 & 0 & 0.5 \\ 0 & 0 & 0.5 & 0 \end{pmatrix}$$

$$L = D - W = \begin{pmatrix} 7 & -5 & -2 & 0 \\ -5 & 9 & -4 & 0 \\ -2 & -4 & 6.5 & -0.5 \\ 0 & 0 & -0.5 & 0.5 \end{pmatrix}$$

Every row and column of $L$ still sums to 0, and
$\mathrm{tr}(L) = \sum_{i=1}^{n} d_i = 2 \sum_{\{i,j\} \in E} w_{ij} = 23$

# Connection with the oriented incidence matrix

We can also factor $L$ as a Gram matrix $B^T B$ by assigning an arbitrary orientation to each edge of $G = (V, E)$

**Definition 159 (Oriented edge-vertex incidence matrix)**

Let $G = (V, E)$ have $n = |V|$ vertices and $m = |E|$ edges $e_1, \ldots, e_m$. Choose an arbitrary direction $e_k = (i, j)$ for each edge $(i \to j)$. The **oriented incidence matrix** $B = [b_{kv}] \in \mathcal{M}_{m,n}(\{-1, 0, 1\})$ has one row per edge and one column per vertex:

$$b_{kv} = \begin{cases} +1 & \text{if } v = i \text{ (tail of } e_k) \\ -1 & \text{if } v = j \text{ (head of } e_k) \\ 0 & \text{otherwise} \end{cases}$$

**Proposition 160 (Incidence factorisation of the Laplacian)**

*For any choice of edge orientations on an unweighted graph $G$, $L = B^T B$*

# Proof of Proposition ?? ($L = B^T B$) and worked check

**Proof:** The $(u, v)$-entry of $B^T B \in \mathcal{M}_n(\mathbb{R})$ is the dot product of columns $u$ and $v$ of $B$:

$$(B^T B)_{uv} = \sum_{k=1}^{m} b_{ku} b_{kv}$$

- If $u = v$: $b_{ku}^2 = (\pm 1)^2 = 1$ whenever edge $e_k$ is incident to vertex $u$, and 0 otherwise, so $(B^T B)_{uu} = \sum_{k=1}^{m} b_{ku}^2 = d_u = \ell_{uu}$
- If $u \neq v$: the product $b_{ku} b_{kv}$ is non-zero only if edge $e_k$ connects $u$ and $v$, in which case one entry is $+1$ and the other is $-1$, so $b_{ku} b_{kv} = -1 = \ell_{uv}$ □

**Check on the 4-vertex paw graph** (orienting $e_1 = (1, 2), e_2 = (1, 3), e_3 = (2, 3), e_4 = (3, 4)$):

$$B = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} \implies B^T B = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} = L$$

# Elementary matrix properties of $L$

From $L = D - A$ and $L = B^T B$, several useful matrix identities follow immediately:

1. **Trace of $L$:** By the handshaking lemma, the sum of the eigenvalues of $L$ equals twice the total edge weight:

$$\sum_{k=1}^{n} \lambda_k = \operatorname{tr}(L) = \sum_{i=1}^{n} d_i = 2 \sum_{\{i,j\} \in E} a_{ij} \qquad (= 2|E| \text{ for an unweighted graph})$$

2. **Sum of rank-one edge matrices:** Each edge $e = \{i, j\} \in E$ contributes a $2 \times 2$ sub-Laplacian on vertices $i$ and $j$. Letting $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$ denote the standard basis of $\mathbb{R}^n$,

$$L = \sum_{\{i,j\} \in E} a_{ij} (\boldsymbol{e}_i - \boldsymbol{e}_j)(\boldsymbol{e}_i - \boldsymbol{e}_j)^T$$

3. **Weak diagonal dominance:** Every row $i$ satisfies $|\ell_{ii}| = d_i = \sum_{j \neq i} a_{ij} = \sum_{j \neq i} |\ell_{ij}|$

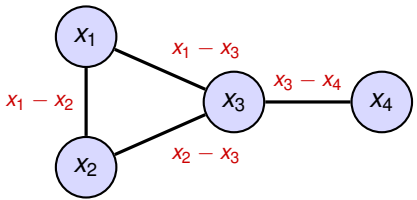
# Vectors on $\mathbb{R}^n$ as functions on the vertices

How does $L$ act on a vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)^T \in \mathbb{R}^n$?

Think of $\boldsymbol{x} \in \mathbb{R}^n$ as assigning a real number (a "signal" or "potential") $x_i \in \mathbb{R}$ to each vertex $i \in V$:

- At vertex $i$, the $i$th entry of $L\boldsymbol{x}$ is

$$(L\boldsymbol{x})_i = d_i x_i - \sum_{j=1}^{n} a_{ij} x_j = \sum_{j=1}^{n} a_{ij}(x_i - x_j)$$




- Along each edge $e_k = (i, j)$, the $k$th entry of $B\boldsymbol{x}$ is the **edge difference**

$$(B\boldsymbol{x})_k = x_i - x_j$$

Multiplying on the left by $\boldsymbol{x}^T$ produces the fundamental identity of spectral graph theory

# The Laplacian quadratic form

**Theorem 161 (Laplacian quadratic form)**

*Let $G = (V, E)$ be an undirected graph on $n$ vertices with symmetric weight matrix $A = [a_{ij}]$ ($a_{ij} = a_{ji} \geq 0$) and combinatorial Laplacian $L = D - A$. Then for every vector $\boldsymbol{x} = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$,*

$$\boldsymbol{x}^T L \boldsymbol{x} = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}(x_i - x_j)^2 = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2$$

*In particular, for an unweighted simple graph ($a_{ij} \in \{0, 1\}$),*

$$\boldsymbol{x}^T L \boldsymbol{x} = \sum_{\{i,j\} \in E} (x_i - x_j)^2$$

# Proof of Theorem ?? (Method 1: direct algebraic expansion)

Let $\boldsymbol{x} = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$. Using $L = D - A$ and $d_i = \sum_{j=1}^{n} a_{ij}$, we expand

$$\boldsymbol{x}^T L \boldsymbol{x} = \boldsymbol{x}^T D \boldsymbol{x} - \boldsymbol{x}^T A \boldsymbol{x} = \sum_{i=1}^{n} d_i x_i^2 - \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j$$

$$= \sum_{i=1}^{n} \left( \sum_{j=1}^{n} a_{ij} \right) x_i^2 - \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j$$

By symmetry $a_{ij} = a_{ji}$, relabelling dummy indices $i \leftrightarrow j$ gives $\sum_{i,j} a_{ij} x_i^2 = \sum_{i,j} a_{ji} x_j^2 = \sum_{i,j} a_{ij} x_j^2$, so we can split the first sum into two equal halves:

$$\boldsymbol{x}^T L \boldsymbol{x} = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i^2 - \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j + \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_j^2$$

$$= \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} \left( x_i^2 - 2 x_i x_j + x_j^2 \right) = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} (x_i - x_j)^2$$

Since each undirected edge $\{i, j\} \in E$ appears twice in the double sum (as $(i, j)$

# Proof of Theorem ?? (Method 2: via the incidence matrix)

For an unweighted graph $G = (V, E)$, Proposition **??** gives $L = B^T B$, where $B \in \mathcal{M}_{|E|,n}$ is the oriented incidence matrix

For any $\boldsymbol{x} \in \mathbb{R}^n$, the $k$th entry of $B\boldsymbol{x} \in \mathbb{R}^{|E|}$ corresponding to oriented edge $e_k = (i, j)$ is

$$(B\boldsymbol{x})_k = x_i - x_j$$

Therefore,

$$\boldsymbol{x}^T L\boldsymbol{x} = \boldsymbol{x}^T(B^T B)\boldsymbol{x} = (B\boldsymbol{x})^T(B\boldsymbol{x}) = \|B\boldsymbol{x}\|_2^2 = \sum_{k=1}^{|E|}(B\boldsymbol{x})_k^2 = \sum_{\{i,j\}\in E}(x_i - x_j)^2$$

(For a weighted graph with diagonal edge-weight matrix $W_E = \mathrm{diag}(w_{e_1}, \ldots, w_{e_{|E|}})$, writing $L = B^T W_E B = (W_E^{1/2} B)^T (W_E^{1/2} B)$ gives $\boldsymbol{x}^T L\boldsymbol{x} = \|W_E^{1/2} B\boldsymbol{x}\|_2^2 = \sum_{\{i,j\}\in E} w_{ij}(x_i - x_j)^2$)

# Geometric interpretation: smoothness (Dirichlet energy) on a graph

The quadratic form $\boldsymbol{x}^T L \boldsymbol{x} = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2$ is called the **Dirichlet energy** of the vector $\boldsymbol{x}$ on the graph $G$:

- Every edge $\{i, j\} \in E$ acts like a spring with stiffness $a_{ij}$ that penalises the squared difference $(x_i - x_j)^2$ between its endpoints
- **Smooth vector** ($\boldsymbol{x}^T L \boldsymbol{x}$ small): adjacent vertices have nearly equal values ($x_i \approx x_j$ whenever $a_{ij}$ is large)
- **Constant vector** ($\boldsymbol{x}^T L \boldsymbol{x} = 0$): $x_i = x_j$ across every edge of $G$
- **Rough / oscillating vector** ($\boldsymbol{x}^T L \boldsymbol{x}$ large): adjacent vertices have very different values (e.g. alternating signs $+1$ and $-1$ across edges)

# Example: evaluating $\boldsymbol{x}^T L\boldsymbol{x}$ on the 4-vertex paw graph

Take the 4-vertex paw graph with edges $\{1,2\},\{1,3\},\{2,3\},\{3,4\}$ and compare two vectors of equal norm:

**Case 1 (smooth):** $\boldsymbol{x} = (1,1,0,-1)^T$

$$\begin{aligned}\boldsymbol{x}^T L\boldsymbol{x} &= (x_1-x_2)^2 + (x_1-x_3)^2 \\ &\quad + (x_2-x_3)^2 + (x_3-x_4)^2 \\ &= (1-1)^2 + (1-0)^2 \\ &\quad + (1-0)^2 + (0-(-1))^2 \\ &= 0+1+1+1 = 3\end{aligned}$$

Adjacent vertices $\{1,2\}$ share the same value 1, and values change gradually along the graph

**Case 2 (oscillating):** $\boldsymbol{y} = (1,-1,0,1)^T$

$$\begin{aligned}\boldsymbol{y}^T L\boldsymbol{y} &= (y_1-y_2)^2 + (y_1-y_3)^2 \\ &\quad + (y_2-y_3)^2 + (y_3-y_4)^2 \\ &= (1-(-1))^2 + (1-0)^2 \\ &\quad + (-1-0)^2 + (0-1)^2 \\ &= 4+1+1+1 = 7\end{aligned}$$

Adjacent vertices 1 and 2 jump from $+1$ to $-1$, producing a much larger energy

# Fundamental spectral properties of the Laplacian

**Theorem 162 (Fundamental spectral properties of $L$)**

*Let $G = (V, E)$ be an undirected graph on $n$ vertices with symmetric nonnegative weight matrix $A = [a_{ij}]$ and combinatorial Laplacian $L = D - A$. Then:*

1. *$L$ is real and symmetric ($L^T = L$), so it has $n$ real eigenvalues and an orthonormal basis of real eigenvectors $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$*
2. *$L$ is **positive semi-definite** ($\boldsymbol{x}^T L \boldsymbol{x} \geq 0$ for all $\boldsymbol{x} \in \mathbb{R}^n$), so all eigenvalues of $L$ are nonnegative and can be ordered in ascending order:*

$$0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$$

3. *The all-ones vector $\mathbb{1} = (1, 1, \ldots, 1)^T$ satisfies $L\mathbb{1} = \mathbf{0}$, so $\lambda_1 = 0$ is always the smallest eigenvalue of $L$ with eigenvector $\boldsymbol{v}_1 = \mathbb{1}$ (or unit eigenvector $\frac{1}{\sqrt{n}}\mathbb{1}$)*

# Proof of Theorem ??

1. **Symmetry and real spectrum:** Since $D$ is diagonal ($D^T = D$) and $A$ is symmetric ($A^T = A$),
$$L^T = (D - A)^T = D^T - A^T = D - A = L$$
By the spectral theorem for real symmetric matrices, $L$ has $n$ real eigenvalues and an orthonormal basis of real eigenvectors in $\mathbb{R}^n$
2. **Positive semi-definiteness and $\lambda_k \geq 0$:** By Theorem **??**, for any $\boldsymbol{x} \in \mathbb{R}^n$,
$$\boldsymbol{x}^T L \boldsymbol{x} = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2 \geq 0$$
since $a_{ij} \geq 0$ and $(x_i - x_j)^2 \geq 0$. If $L\boldsymbol{v}_k = \lambda_k \boldsymbol{v}_k$ with $\boldsymbol{v}_k \neq \mathbf{0}$, then
$$\lambda_k = \frac{\boldsymbol{v}_k^T (L\boldsymbol{v}_k)}{\boldsymbol{v}_k^T \boldsymbol{v}_k} = \frac{\boldsymbol{v}_k^T L \boldsymbol{v}_k}{\|\boldsymbol{v}_k\|_2^2} \geq 0$$
3. **Zero row sums ($L\mathbb{1} = \mathbf{0}$):** For each row $i = 1, \ldots, n$,
$$(L\mathbb{1})_i = \sum_{j=1}^{n} \ell_{ij} \cdot 1 = d_i - \sum_{j=1}^{n} a_{ij} = d_i - d_i = 0 \implies L\mathbb{1} = \mathbf{0} = 0 \cdot \mathbb{1}$$ □

# Beware of the eigenvalue ordering convention!

In PCA (Unit 05) and Perron-Frobenius theory (Units 08–09), we ordered eigenvalues from *largest to smallest* because we wanted maximum variance or the dominant long-term state

For the graph Laplacian $L$, the universal convention is the **ascending order**

$$0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$$

Why? Because for any unit eigenvector $\boldsymbol{v}_k$ ($\|\boldsymbol{v}_k\|_2 = 1$),

$$\lambda_k = \boldsymbol{v}_k^T L \boldsymbol{v}_k = \sum_{\{i,j\} \in E} a_{ij}(v_{ki} - v_{kj})^2$$

- **Smallest eigenvalues ($\lambda_1 = 0, \lambda_2, \ldots$)**: eigenvectors vary *least* across edges, revealing connected components and large-scale community structure
- **Largest eigenvalues ($\ldots, \lambda_n$)**: eigenvectors oscillate rapidly across edges

# Multiplicity of $\lambda = 0$ and connected components

We know $\lambda_1 = 0$ is always an eigenvalue of $L$. When is $\lambda_2 = 0$ as well?

**Theorem 163 (Multiplicity of 0 equals the number of connected components)**

*Let $G = (V, E)$ be an undirected graph with nonnegative edge weights and Laplacian $L$. The multiplicity of* 0 *as an eigenvalue of $L$ (that is, $\text{nullity}(L) = \dim(\text{null}(L))$) equals the number $k$ of connected components $C_1, C_2, \ldots, C_k$ of $G$*

*Moreover, the component indicator vectors $\mathbb{1}_{C_1}, \mathbb{1}_{C_2}, \ldots, \mathbb{1}_{C_k} \in \{0,1\}^n$, defined by*

$$(\mathbb{1}_{C_r})_i = \begin{cases} 1 & \text{if } i \in C_r \\ 0 & \text{otherwise} \end{cases}$$

*form an orthogonal basis of* $\text{null}(L)$

# Proof of Theorem ?? – Part 1: connected graph ($k = 1$)

Suppose first that $G$ is connected ($k = 1$). We must show that $\text{null}(L) = \text{span}\{\mathbb{1}\}$, so $\dim(\text{null}(L)) = 1$

1. We already know $L\mathbb{1} = \mathbf{0}$, so $\text{span}\{\mathbb{1}\} \subseteq \text{null}(L)$
2. Conversely, let $\boldsymbol{x} \in \text{null}(L)$, so $L\boldsymbol{x} = \mathbf{0}$. Taking the dot product with $\boldsymbol{x}$ and applying Theorem **??** gives
$$0 = \boldsymbol{x}^T\mathbf{0} = \boldsymbol{x}^T L\boldsymbol{x} = \sum_{\{i,j\}\in E} a_{ij}(x_i - x_j)^2$$
3. For every edge $\{i, j\} \in E$, we have $a_{ij} > 0$ and $(x_i - x_j)^2 \geq 0$. A sum of nonnegative terms equals 0 if and only if every term is 0:
$$(x_i - x_j)^2 = 0 \implies x_i = x_j \quad \text{for every edge } \{i, j\} \in E$$
4. Since $G$ is connected, any two vertices $u, v \in V$ are linked by a path $u = i_0, i_1, \ldots, i_m = v$ in $G$, so $x_u = x_{i_1} = \cdots = x_v$
5. Hence all entries of $\boldsymbol{x}$ are equal ($x_1 = \cdots = x_n = c$), i.e. $\boldsymbol{x} = c\,\mathbb{1} \in \text{span}\{\mathbb{1}\}$

□

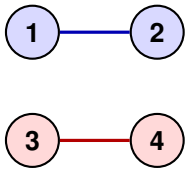
# Proof of Theorem ?? – Part 2: $k$ connected components ($k \geq 2$)

Now suppose $G$ has $k$ connected components $C_1, \ldots, C_k$. Reorder the vertices so that those in $C_1$ come first, followed by $C_2, \ldots, C_k$

1. Because there are no edges between different components, $A$, $D$ and $L$ are block-diagonal:
$$L = \begin{pmatrix} L_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & L_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & L_k \end{pmatrix}$$
where each $L_r$ is the Laplacian of the connected subgraph on $C_r$
2. Writing $\boldsymbol{x} = (\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(k)})^T$ according to the blocks, $L\boldsymbol{x} = \mathbf{0} \iff L_r \boldsymbol{x}^{(r)} = \mathbf{0}$ for every $r = 1, \ldots, k$
3. By Part 1 applied to each connected component $C_r$, $L_r \boldsymbol{x}^{(r)} = \mathbf{0} \iff \boldsymbol{x}^{(r)} = c_r \mathbb{1}_{|C_r|}$, which in $\mathbb{R}^n$ means
$$\boldsymbol{x} = c_1 \mathbb{1}_{C_1} + c_2 \mathbb{1}_{C_2} + \cdots + c_k \mathbb{1}_{C_k}$$
4. Since $C_1, \ldots, C_k$ are disjoint, $\mathbb{1}_{C_r}^T \mathbb{1}_{C_s} = 0$ for $r \neq s$, so $\{\mathbb{1}_{C_1}, \ldots, \mathbb{1}_{C_k}\}$ is an orthogonal basis of $\text{null}(L)$ and $\dim(\text{null}(L)) = k$

# Example: a disconnected graph on 4 vertices

Consider $G = (V, E)$ on $n = 4$ vertices with only two edges $\{1, 2\}$ and $\{3, 4\}$, so $G$ has $k = 2$ connected components $C_1 = \{1, 2\}$ and $C_2 = \{3, 4\}$:




$$L = \left(\begin{array}{cc|cc} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{array}\right) = \begin{pmatrix} L_1 & \mathbf{0} \\ \mathbf{0} & L_2 \end{pmatrix}$$

- Each $2 \times 2$ block $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ has trace 2 and determinant 0, so its eigenvalues are 0 and 2
- The spectrum of $L$ is therefore $\lambda_1 = 0,\ \lambda_2 = 0,\ \lambda_3 = 2,\ \lambda_4 = 2$ (multiplicity of 0 is $k = 2$)
- The orthogonal indicator vectors for null($L$) are

$$\mathbb{1}_{C_1} = (1, 1, 0, 0)^T \qquad \text{and} \qquad \mathbb{1}_{C_2} = (0, 0, 1, 1)^T$$

# Counting connected components with $\lambda = 0$ in R

Let us build a graph on $n = 8$ vertices consisting of $k = 3$ disconnected components (a triangle $\{1, 2, 3\}$, a path $\{4, 5, 6\}$ and an edge $\{7, 8\}$) and check the eigenvalues of $L$:

```
G_comp = make_graph(edges = c(1,2, 2,3, 1,3,  4,5, 5,6,  7,8),
                    n = 8, directed = FALSE)
L_comp = as.matrix(laplacian_matrix(G_comp))
ev_comp = eigen(L_comp, symmetric = TRUE)
# Eigenvalues of L in ascending order
round(rev(ev_comp$values), 4)

## [1] 0 0 0 1 2 3 3 3

# Number of zero eigenvalues (up to numerical precision)
sum(abs(ev_comp$values) < 1e-10)

## [1] 3
```

Exactly three eigenvalues of $L$ equal 0, matching the $k = 3$ connected

# Spectrum of extreme graphs: empty vs complete graph $K_n$

Comparing two extreme graphs on $n$ vertices illustrates how Laplacian eigenvalues measure connectivity:

**1. Empty graph $E_n$ ($|E| = 0$):**

$$L = \mathbf{0}_n$$

- $n$ isolated vertices ($k = n$ components)
- Spectrum of $L$:

$$\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$$

**2. Complete graph $K_n$ (all pairs connected):**

$$L = (n-1)\mathbb{I}_n - (\mathbb{J}_n - \mathbb{I}_n) = n\mathbb{I}_n - \mathbb{1}\mathbb{1}^T$$

- For $\boldsymbol{v}_1 = \mathbb{1}$:
  $L\mathbb{1} = n\mathbb{1} - n\mathbb{1} = \mathbf{0} \implies \lambda_1 = 0$
- For any $\boldsymbol{x} \perp \mathbb{1}$ ($\mathbb{1}^T\boldsymbol{x} = 0$):

$$L\boldsymbol{x} = n\boldsymbol{x} - \mathbb{1}(\mathbb{1}^T\boldsymbol{x}) = n\boldsymbol{x}$$

- Spectrum of $K_n$: $\lambda_1 = 0$ and $\lambda_2 = \cdots = \lambda_n = n$

For every simple graph $G$ on $n$ vertices, its eigenvalues lie between these two extremes: $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n \leq n$

# What if $G$ is "almost" disconnected?

Suppose $G$ has two dense clusters $C_1$ and $C_2$ that are completely disconnected ($k = 2$):

- Then $\lambda_1 = 0$ and $\lambda_2 = 0$, and any vector in null($L$) is constant on $C_1$ and constant on $C_2$:

$$\boldsymbol{v}_1 = (1, \ldots, 1,\ 1, \ldots, 1)^T, \qquad \boldsymbol{v}_2 = (1, \ldots, 1,\ -1, \ldots, -1)^T \quad (\text{if } |C_1| = |C_2|)$$

Now perturb $G$ slightly by adding a single weak bridge edge between $C_1$ and $C_2$:

- The graph is now connected ($k = 1$), so $\lambda_1 = 0$ and $\lambda_2 > 0$
- Because eigenvalues and eigenvectors of symmetric matrices vary continuously under small perturbations, $\lambda_2$ will be **positive but very small** ($\lambda_2 \approx 0$)
- The associated eigenvector $\boldsymbol{v}_2 \perp \mathbb{1}$ will remain **nearly constant and positive on** $C_1$ and **nearly constant and negative on** $C_2$!

Next, we make this intuition completely rigorous via **graph cuts**

# Partitions and the cut of a graph

Let $G = (V, E)$ be an undirected graph on $n$ vertices with symmetric edge weights $a_{ij} \geq 0$

A **2-way partition** (or **bipartition**) of $G$ splits $V$ into two disjoint non-empty subsets $S$ and $\bar{S} = V \setminus S$ ($S \cup \bar{S} = V$, $S \cap \bar{S} = \emptyset$)

**Definition 164 (Cut of a partition)**

*The **cut** between $S$ and $\bar{S}$ is the total weight of edges with one endpoint in $S$ and the other endpoint in $\bar{S}$:*

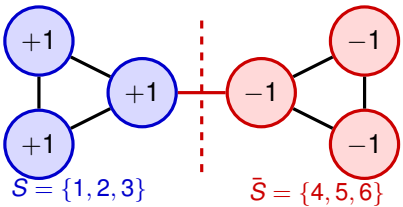
$$\text{cut}(S, \bar{S}) = \sum_{i \in S,\, j \in \bar{S}} a_{ij}$$

*For an unweighted graph ($a_{ij} \in \{0, 1\}$), $\text{cut}(S, \bar{S})$ is simply the number of edges crossing between $S$ and $\bar{S}$*

# Encoding a partition with a $\pm 1$ indicator vector

To connect graph cuts to linear algebra, we represent any bipartition $(S, \bar{S})$ by a **cut indicator vector** $\boldsymbol{x} = (x_1, \ldots, x_n)^T \in \{-1, +1\}^n$:

$$x_i = \begin{cases} +1 & \text{if } i \in S \\ -1 & \text{if } i \in \bar{S} \end{cases}$$




**Proposition 165 (Cut size as a quadratic form)**

*For any partition $(S, \bar{S})$ of $V$ with indicator vector $\boldsymbol{x} \in \{-1, +1\}^n$,*

$$\boldsymbol{x}^T L \boldsymbol{x} = 4 \operatorname{cut}(S, \bar{S})$$

*or equivalently,*

$$\operatorname{cut}(S, \bar{S}) = \frac{1}{4} \boldsymbol{x}^T L \boldsymbol{x}$$

# Proof of Proposition ?? ($\boldsymbol{x}^T L \boldsymbol{x} = 4\,\mathrm{cut}(S, \bar{S})$)

Let $\boldsymbol{x} \in \{-1, +1\}^n$ be the cut indicator vector for $(S, \bar{S})$. For any edge $\{i, j\} \in E$, look at the squared difference $(x_i - x_j)^2$:

- **Both endpoints in** $S$ ($i \in S, j \in S$): $x_i = +1$ and $x_j = +1$, so
$$(x_i - x_j)^2 = (1 - 1)^2 = 0$$
- **Both endpoints in** $\bar{S}$ ($i \in \bar{S}, j \in \bar{S}$): $x_i = -1$ and $x_j = -1$, so
$$(x_i - x_j)^2 = (-1 - (-1))^2 = 0$$
- **Endpoints in different sets** ($i \in S, j \in \bar{S}$ or $i \in \bar{S}, j \in S$): $x_i - x_j = \pm 2$, so
$$(x_i - x_j)^2 = (\pm 2)^2 = 4$$

Substituting into Theorem ?? ($\boldsymbol{x}^T L \boldsymbol{x} = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2$), only the edges crossing between $S$ and $\bar{S}$ survive, each multiplied by 4:

$$\boldsymbol{x}^T L \boldsymbol{x} = \sum_{i \in S,\, j \in \bar{S}} 4\, a_{ij} = 4 \sum_{i \in S,\, j \in \bar{S}} a_{ij} = 4\,\mathrm{cut}(S, \bar{S}) \qquad \square$$

# Example: comparing two cuts on the 4-vertex paw graph

Recall the 4-vertex paw graph with Laplacian

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Let us compare two candidate partitions using $\text{cut}(S, \bar{S}) = \frac{1}{4}\boldsymbol{x}^T L \boldsymbol{x}$:

**Partition A:** $S_1 = \{1, 2, 3\}$, $\bar{S}_1 = \{4\}$

$$\boldsymbol{x}^{(1)} = (1, 1, 1, -1)^T$$

$$L\boldsymbol{x}^{(1)} = \begin{pmatrix} 0 \\ 0 \\ 2 \\ -2 \end{pmatrix} \implies (\boldsymbol{x}^{(1)})^T L \boldsymbol{x}^{(1)} = 4$$

$$\text{cut}(S_1, \bar{S}_1) = \frac{4}{4} = 1 \quad \text{(only edge } \{3, 4\}\text{)}$$

**Partition B:** $S_2 = \{1, 3\}$, $\bar{S}_2 = \{2, 4\}$

$$\boldsymbol{x}^{(2)} = (1, -1, 1, -1)^T$$

$$L\boldsymbol{x}^{(2)} = \begin{pmatrix} 2 \\ -4 \\ 4 \\ -2 \end{pmatrix} \implies (\boldsymbol{x}^{(2)})^T L \boldsymbol{x}^{(2)} = 12$$

$$\text{cut}(S_2, \bar{S}_2) = \frac{12}{4} = 3 \quad \text{(edges } \{1, 2\}, \{2, 3\},$$

# Why unconstrained minimum cut is not enough

Since $\text{cut}(S, \bar{S}) = \frac{1}{4}\boldsymbol{x}^T L \boldsymbol{x}$, can we cluster a graph simply by minimising $\boldsymbol{x}^T L \boldsymbol{x}$ over $\boldsymbol{x} \in \{-1, +1\}^n$?

1. **Trivial all-in-one solution:** Without constraints, the minimum of $\boldsymbol{x}^T L \boldsymbol{x}$ over $\{-1, +1\}^n$ is 0, achieved at $\boldsymbol{x} = +\mathbb{1}$ ($S = V, \bar{S} = \emptyset$) and $\boldsymbol{x} = -\mathbb{1}$ ($S = \emptyset, \bar{S} = V$)

2. **Unbalanced "whisker" cuts:** Even if we forbid $\boldsymbol{x} = \pm\mathbb{1}$, minimum cut often slices off a single degree-1 vertex (like vertex 4 in the paw graph) rather than splitting the graph into two meaningful communities of comparable size

To find balanced communities, we must add a **balance constraint** requiring $|S|$ and $|\bar{S}|$ to be equal (or comparable) in size

# Encoding balanced bisection algebraically

Suppose $n$ is even and we want a **balanced bisection** with $|S| = |\bar{S}| = n/2$

For any cut indicator vector $\boldsymbol{x} \in \{-1, +1\}^n$, notice two algebraic facts:

1. **Balance equals orthogonality to $\mathbb{1}$:** Since $x_i = +1$ on $S$ and $x_i = -1$ on $\bar{S}$,

$$\mathbb{1}^T \boldsymbol{x} = \sum_{i=1}^{n} x_i = |S| - |\bar{S}|$$

Therefore,

$$|S| = |\bar{S}| \quad \Longleftrightarrow \quad \mathbb{1}^T \boldsymbol{x} = 0 \quad \Longleftrightarrow \quad \boldsymbol{x} \perp \mathbb{1}$$

2. **Fixed Euclidean norm:** Every vector $\boldsymbol{x} \in \{-1, +1\}^n$ has squared norm

$$\|\boldsymbol{x}\|_2^2 = \boldsymbol{x}^T \boldsymbol{x} = \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} (\pm 1)^2 = n$$

# General balanced cuts: RatioCut for unequal cluster sizes

What if the two true clusters do not have the exact same size ($|S| \neq |\bar{S}|$)? We penalise small sets by minimising the **RatioCut** (Hagen & Kahng, 1992):

$$\text{RatioCut}(S, \bar{S}) = \text{cut}(S, \bar{S}) \left( \frac{1}{|S|} + \frac{1}{|\bar{S}|} \right) = \frac{n}{|S|\,|\bar{S}|} \text{cut}(S, \bar{S})$$

Define the weighted indicator vector $\boldsymbol{x} = (x_1, \ldots, x_n)^T$ by

$$x_i = \begin{cases} +\sqrt{|\bar{S}|/|S|} & \text{if } i \in S \\ -\sqrt{|S|/|\bar{S}|} & \text{if } i \in \bar{S} \end{cases}$$

Then one checks directly that:

1. $\mathbb{1}^T \boldsymbol{x} = |S|\sqrt{|\bar{S}|/|S|} - |\bar{S}|\sqrt{|S|/|\bar{S}|} = \sqrt{|S|\,|\bar{S}|} - \sqrt{|S|\,|\bar{S}|} = 0$ ($\boldsymbol{x} \perp \mathbb{1}$)
2. $\|\boldsymbol{x}\|_2^2 = |S|\frac{|\bar{S}|}{|S|} + |\bar{S}|\frac{|S|}{|\bar{S}|} = |\bar{S}| + |S| = n$, and $\boldsymbol{x}^T L \boldsymbol{x} = n\,\text{RatioCut}(S, \bar{S})$

# From NP-hard discrete cut to continuous spectral relaxation

Both balanced minimum bisection and RatioCut lead to minimising $\boldsymbol{x}^T L \boldsymbol{x}$ subject to $\boldsymbol{x} \perp \mathbb{1}$, $\|\boldsymbol{x}\|_2^2 = n$ and a discrete two-value constraint on the entries $x_i$:

$$\min_{\substack{\boldsymbol{x} \in \{-1,+1\}^n \\ \boldsymbol{x}^T \mathbb{1} = 0}} \frac{1}{4} \boldsymbol{x}^T L \boldsymbol{x}$$

- **Computational barrier:** Checking all balanced partitions $\binom{n}{n/2}$ is **NP-hard** (for $n = 100$ vertices, $\binom{100}{50} \approx 1.0 \times 10^{29}$)
- **Spectral relaxation:** Relax the discrete constraint on $x_i$ to allow *arbitrary real values* $\boldsymbol{x} \in \mathbb{R}^n$ that still satisfy the norm constraint $\boldsymbol{x}^T \boldsymbol{x} = n$ and the balance constraint $\boldsymbol{x}^T \mathbb{1} = 0$:

$$\min_{\substack{\boldsymbol{x} \in \mathbb{R}^n \\ \boldsymbol{x}^T \boldsymbol{x} = n,\ \boldsymbol{x}^T \mathbb{1} = 0}} \boldsymbol{x}^T L \boldsymbol{x} \quad \Longleftrightarrow \quad \min_{\substack{\boldsymbol{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\} \\ \boldsymbol{x} \perp \mathbb{1}}} \frac{\boldsymbol{x}^T L \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}}$$

# Solving the relaxed problem via the Rayleigh quotient

**Theorem 166 (Rayleigh quotient theorem for $\lambda_2$)**

*Let $L$ be the Laplacian of an undirected graph $G$ on $n$ vertices with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and orthonormal eigenvectors $\boldsymbol{v}_1 = \frac{1}{\sqrt{n}}\mathbb{1}, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$. Then*

$$\min_{\substack{\boldsymbol{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\} \\ \boldsymbol{x} \perp \mathbb{1}}} \frac{\boldsymbol{x}^T L \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} = \lambda_2$$

*and the minimum is attained when $\boldsymbol{x}$ is any non-zero scalar multiple of $\boldsymbol{v}_2$*

In particular, if we scale $\boldsymbol{v}_2$ so that $\|\boldsymbol{x}\|_2^2 = n$, the optimal continuous vector is $\boldsymbol{x} = \sqrt{n}\,\boldsymbol{v}_2$, with objective value $\boldsymbol{x}^T L \boldsymbol{x} = n\lambda_2$ (giving the lower bound $\text{cut}(S, \bar{S}) \geq \frac{n}{4}\lambda_2$ for any balanced bisection!)

# Proof of Theorem ??

Let $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ be an orthonormal basis of eigenvectors of $L$ with $L\boldsymbol{v}_k = \lambda_k \boldsymbol{v}_k$, where $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and $\boldsymbol{v}_1 = \frac{1}{\sqrt{n}}\mathbb{1}$

1. Expand any vector $\boldsymbol{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ in the orthonormal basis $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$:

$$\boldsymbol{x} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_n\boldsymbol{v}_n, \qquad c_k = \boldsymbol{v}_k^T\boldsymbol{x}$$

2. The orthogonality constraint $\boldsymbol{x} \perp \mathbb{1}$ means $c_1 = \boldsymbol{v}_1^T\boldsymbol{x} = \frac{1}{\sqrt{n}}\mathbb{1}^T\boldsymbol{x} = 0$, so

$$\boldsymbol{x} = \sum_{k=2}^{n} c_k\boldsymbol{v}_k \qquad \text{and} \qquad \boldsymbol{x}^T\boldsymbol{x} = \sum_{k=2}^{n} c_k^2 > 0$$

3. Since $L\boldsymbol{x} = \sum_{k=2}^{n} c_k\lambda_k\boldsymbol{v}_k$ and the $\boldsymbol{v}_k$ are orthonormal,

$$\frac{\boldsymbol{x}^T L\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}} = \frac{\sum_{k=2}^{n}\lambda_k c_k^2}{\sum_{k=2}^{n} c_k^2} \geq \frac{\sum_{k=2}^{n}\lambda_2 c_k^2}{\sum_{k=2}^{n} c_k^2} = \lambda_2$$

with equality when $c_2 \neq 0$ and $c_3 = \cdots = c_n = 0$, i.e. $\boldsymbol{x} = c_2\boldsymbol{v}_2$ □

# Algebraic connectivity and the Fiedler vector

Miroslav Fiedler (1973) discovered the fundamental role of $\lambda_2$ and $\boldsymbol{v}_2$ in measuring how well-connected a graph is

**Definition 167 (Algebraic connectivity and Fiedler vector)**

*Let $G = (V, E)$ be an undirected graph on $n$ vertices with Laplacian eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$*

- *The second smallest eigenvalue $\lambda_2$ is called the **algebraic connectivity** (or **Fiedler value**) of $G$*
- *An eigenvector $\boldsymbol{v}_2$ associated with $\lambda_2$ (orthogonal to $\mathbb{1}$) is called a **Fiedler vector** of $G$*

- By Theorem **??**, $\lambda_2 > 0$ **if and only if** *G* **is connected**
- Because $\boldsymbol{v}_2 \perp \mathbb{1}$, the entries of $\boldsymbol{v}_2$ sum to 0 ($\sum_{i=1}^{n} v_{2i} = 0$), so $\boldsymbol{v}_2$ always has **both positive and negative entries**!

# The 2-way spectral partitioning algorithm

How do we turn the continuous Fiedler vector $\boldsymbol{v}_2 = (v_{21}, \ldots, v_{2n})^T \in \mathbb{R}^n$ back into a discrete partition $(S, \bar{S})$? We round its entries by their signs!

## Algorithm: 2-way unnormalised spectral clustering (Fiedler bisection)

**Input:** Undirected graph $G = (V, E)$ with symmetric adjacency/similarity matrix $A \in \mathcal{M}_n(\mathbb{R}_{\geq 0})$

1. Compute the degree matrix $D = \operatorname{diag}(A\mathbb{1})$ and the Laplacian $L = D - A$
2. Compute the second smallest eigenvalue $\lambda_2$ and associated Fiedler vector $\boldsymbol{v}_2 = (v_{21}, \ldots, v_{2n})^T$ of $L$
3. Partition the vertices according to the sign of $v_{2i}$:

$$S = \{i \in V : v_{2i} \geq 0\}, \qquad \bar{S} = \{i \in V : v_{2i} < 0\}$$

(Alternatively, split at the median of $\boldsymbol{v}_2$ if an exact $n/2 - n/2$ split is required)

Note: Because $-\boldsymbol{v}_2$ is also a Fiedler vector, flipping the sign of $\boldsymbol{v}_2$ simply swaps the labels $S \leftrightarrow \bar{S}$, which defines the *exact same* partition of $V$

# Extension to $k$ clusters and normalised Laplacians

What if a graph (or dataset) has $k > 2$ clusters, or vertices with wildly different degrees?

- **$k$-way spectral clustering:**
  1. Compute the first $k$ eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k$ of $L$ associated with the $k$ smallest eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_k$
  2. Stack them as columns of the matrix $U = [\boldsymbol{v}_1 \;\; \boldsymbol{v}_2 \;\; \cdots \;\; \boldsymbol{v}_k] \in \mathcal{M}_{n,k}(\mathbb{R})$
  3. Treat row $i$ of $U$ as the new embedding $\boldsymbol{y}_i \in \mathbb{R}^k$ of vertex $i$, and run $k$-means on $\{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n\} \subset \mathbb{R}^k$
- **Normalised graph Laplacians** (used when degrees $d_i$ vary over orders of magnitude):

$$L_{\text{sym}} = D^{-1/2} L D^{-1/2} = \mathbb{I} - D^{-1/2} A D^{-1/2}, \qquad L_{\text{rw}} = D^{-1} L = \mathbb{I} - D^{-1} A$$

Notice that $D^{-1}A$ is the random-walk transition matrix on $G$! Spectral clustering with $L_{\text{sym}}$ or $L_{\text{rw}}$ relaxes the degree-normalised cut (Shi & Malik 2000; Ng, Jordan & Weiss 2001)

# Choosing the number of clusters $k$: the eigengap heuristic

How do we choose the number of clusters $k$ from the spectrum $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$?

- By Theorem **??**, if $G$ consists of $k$ completely disconnected clusters, then

$$\lambda_1 = \lambda_2 = \cdots = \lambda_k = 0 \qquad \text{and} \qquad \lambda_{k+1} > 0$$

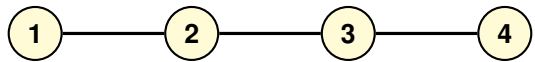
- When those $k$ dense clusters are joined by only a few weak between-cluster edges, the first $k$ eigenvalues remain **very close to** 0, while $\lambda_{k+1}$ jumps sharply upward
- **Eigengap heuristic**: plot $\lambda_1, \lambda_2, \ldots, \lambda_m$ in ascending order and look for the index $k$ with the largest **eigengap**

$$\Delta_k = \lambda_{k+1} - \lambda_k$$

For example, if $\lambda_1 = 0,\ \lambda_2 = 0.04,\ \lambda_3 = 0.07,\ \lambda_4 = 2.85,\ \lambda_5 = 3.10$, the large gap $\Delta_3 = \lambda_4 - \lambda_3 = 2.78$ strongly indicates $k = 3$ clusters

# Example 1: spectral bisection of the 4-vertex path $P_4$

Consider the 4-vertex path graph $P_4$:




**Step 1 – Degree, adjacency and Laplacian matrices:**

$$D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \implies L = D - A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

# Example 1 (continued): Fiedler vector and cut of $P_4$

**Step 2 – Verify the Fiedler eigenpair $(\lambda_2, \boldsymbol{v}_2)$:** Claim: $\lambda_2 = 2 - \sqrt{2} \approx 0.586$ and $\boldsymbol{v}_2 = \begin{pmatrix} 1, & \sqrt{2}-1, & 1-\sqrt{2}, & -1 \end{pmatrix}^T \approx \begin{pmatrix} 1, & 0.414, & -0.414, & -1 \end{pmatrix}^T$
Let us verify $L\boldsymbol{v}_2 = \lambda_2 \boldsymbol{v}_2$ directly:

$$L\boldsymbol{v}_2 = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ \sqrt{2}-1 \\ 1-\sqrt{2} \\ -1 \end{pmatrix} = \begin{pmatrix} 2-\sqrt{2} \\ 3\sqrt{2}-4 \\ 4-3\sqrt{2} \\ \sqrt{2}-2 \end{pmatrix} = (2-\sqrt{2}) \begin{pmatrix} 1 \\ \sqrt{2}-1 \\ 1-\sqrt{2} \\ -1 \end{pmatrix} = \lambda_2 \boldsymbol{v}_2$$

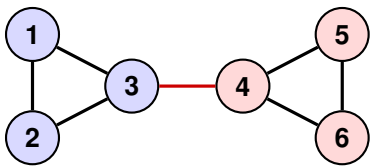
(since $(2-\sqrt{2})(\sqrt{2}-1) = 3\sqrt{2}-4$). Notice also that
$\mathbb{1}^T \boldsymbol{v}_2 = 1 + (\sqrt{2}-1) + (1-\sqrt{2}) - 1 = 0$

**Step 3 – Partition by the signs of $\boldsymbol{v}_2$:**

- $v_{21} = 1 > 0$ and $v_{22} = \sqrt{2} - 1 \approx 0.414 > 0 \implies S = \{1, 2\}$
- $v_{23} = 1 - \sqrt{2} \approx -0.414 < 0$ and $v_{24} = -1 < 0 \implies \bar{S} = \{3, 4\}$
- The spectral cut severing the middle edge $\{2, 3\}$ has $\text{cut}(S, \bar{S}) = 1$

# Example 2: the 6-vertex barbell graph

Consider two triangles $C_1 = \{1, 2, 3\}$ and $C_2 = \{4, 5, 6\}$ joined by a single bridge edge $\{3, 4\}$:




Degrees: $d_1 = d_2 = d_5 = d_6 = 2$, $d_3 = d_4 = 3$

$$L = \left(\begin{array}{ccc|ccc} 2 & -1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ -1 & -1 & 3 & -1 & 0 & 0 \\ \hline 0 & 0 & -1 & 3 & -1 & -1 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & -1 & 2 \end{array}\right)$$

Notice that $L$ is nearly block-diagonal: the two $3 \times 3$ blocks are coupled only through $\ell_{34} = \ell_{43} = -1$ and the extra $+1$ on $\ell_{33}$ and $\ell_{44}$

# Example 2 (continued): exact Fiedler vector of the barbell

By symmetry between $\{1,2\}$ and $\{5,6\}$ and antisymmetry across the bridge $\{3,4\}$ (so that $\mathbb{1}^T \boldsymbol{v}_2 = 0$), we look for a Fiedler vector of the form

$$\boldsymbol{v}_2 = (1,\ 1,\ c,\ -c,\ -1,\ -1)^T$$

Substitute into $L\boldsymbol{v}_2 = \lambda \boldsymbol{v}_2$:

- **Row 1 (and Row 2):** $2(1) - 1(1) - 1(c) = 1 - c = \lambda \implies c = 1 - \lambda$
- **Row 3:** $-1(1) - 1(1) + 3c - 1(-c) = 4c - 2 = \lambda c$

Substitute $c = 1 - \lambda$ into $4c - 2 = \lambda c$:

$$4(1-\lambda) - 2 = \lambda(1-\lambda) \iff \lambda^2 - 5\lambda + 2 = 0 \implies \lambda_2 = \frac{5 - \sqrt{17}}{2} \approx 0.4384$$

Then $c = 1 - \lambda_2 = \frac{\sqrt{17}-3}{2} \approx 0.5616 > 0$, giving the Fiedler vector

$$\boldsymbol{v}_2 \approx (1,\ 1,\ 0.562,\ -0.562,\ -1,\ -1)^T$$

Splitting by sign ($\geq 0$ vs $< 0$) gives $S = \{1,2,3\}$ and $\bar{S} = \{4,5,6\}$, cutting only the bridge $\{3,4\}$!

# Computing the barbell Fiedler vector in R

We can build the barbell Laplacian in R (either from $D - A$ or with `igraph::laplacian_matrix`) and inspect its eigenvalues and Fiedler vector:

```
A_barbell = matrix(c(
  0, 1, 1, 0, 0, 0,
  1, 0, 1, 0, 0, 0,
  1, 1, 0, 1, 0, 0,
  0, 0, 1, 0, 1, 1,
  0, 0, 0, 1, 0, 1,
  0, 0, 0, 1, 1, 0
), nrow = 6, byrow = TRUE)
L_barbell = diag(rowSums(A_barbell)) - A_barbell
ev_b = eigen(L_barbell, symmetric = TRUE)
round(rev(ev_b$values), 4)          # eigenvalues in ascending order

## [1] 0.0000 0.4384 3.0000 3.0000 3.0000 4.5616

round(ev_b$vectors[, 5], 4)         # unit Fiedler vector v_2
```

# Real-world benchmark: Zachary's Karate club network

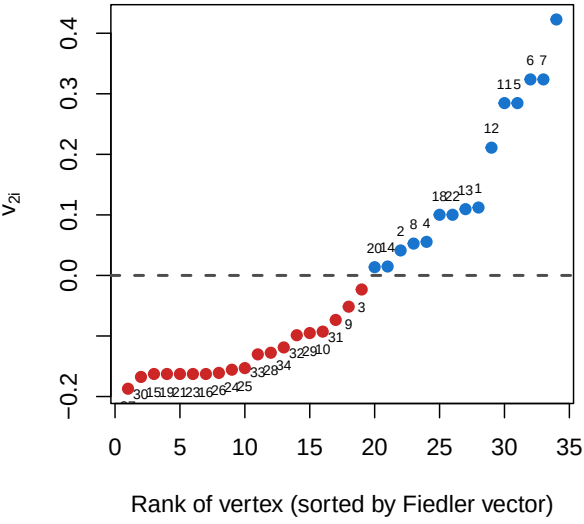
Zachary (1977) recorded social interactions among $n = 34$ members of a university karate club just before a dispute between the instructor (Mr Hi, vertex 1) and the club president (John A, vertex 34) split the club into two factions

```
G_karate = make_graph("Zachary")
L_karate = as.matrix(laplacian_matrix(G_karate))
ev_k = eigen(L_karate, symmetric = TRUE)
n_k = vcount(G_karate)
# Smallest three eigenvalues of L
round(ev_k$values[n_k:(n_k - 2)], 4)

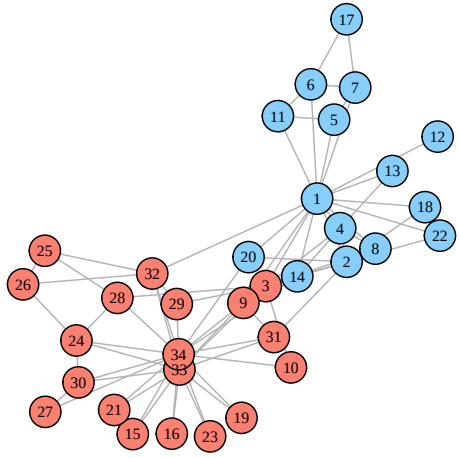
## [1] 0.0000 0.4685 0.9092

# Fiedler vector v_2 (oriented so vertex 1 is positive)
v2_k = ev_k$vectors[, n_k - 1]
if (v2_k[1] < 0) v2_k = -v2_k
table(Sign_of_v2 = ifelse(v2_k >= 0, "Group 1 (+)", "Group 2 (-)"))
```

## Sorted entries of the Fiedler vector




## Karate club partitioned by sign(v2)



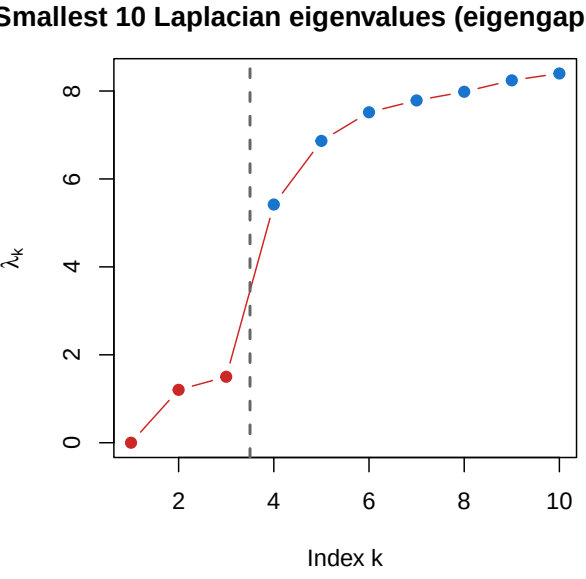
# $k$-way spectral clustering in R ($k = 3$ communities)

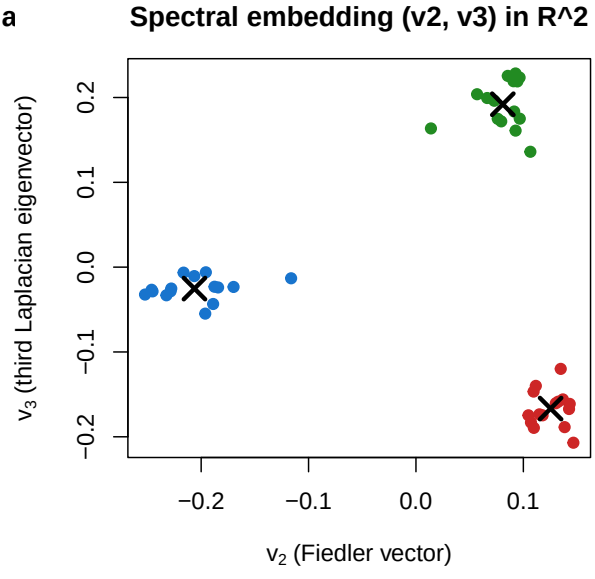
Suppose a network has $k = 3$ dense communities of 15 vertices each ($n = 45$), with high within-community edge probability ($p_{\text{in}} = 0.7$) and low between-community probability ($p_{\text{out}} = 0.03$):

```
set.seed(2740)
pm = matrix(0.03, nrow = 3, ncol = 3); diag(pm) = 0.70
G_sbm = sample_sbm(45, pref.matrix = pm, block.sizes = c(15, 15, 15))
L_sbm = as.matrix(laplacian_matrix(G_sbm))
ev_sbm = eigen(L_sbm, symmetric = TRUE)
# Smallest 6 eigenvalues of L (note the jump after lambda_3!)
round(rev(ev_sbm$values)[1:6], 3)

## [1] 0.000 1.203 1.500 5.418 6.865 7.517

# Spectral embedding using v_2 and v_3, followed by k-means (k = 3)
U_sbm = ev_sbm$vectors[, c(44, 43)]
km_sbm = kmeans(U_sbm, centers = 3, nstart = 20)
table(True_block = rep(1:3, each = 15), Spectral_cluster = km_sbm$cluster)
```


Smallest 10 Laplacian eigenvalues (eigengap a
λk
0
2
4
6
8
2
4
6
8
10
Index k


Spectral embedding (v2, v3) in R^2
v3 (third Laplacian eigenvector)
0.2
0.1
0.0
−0.1
−0.2
−0.2
−0.1
0.0
0.1
v2 (Fiedler vector)

# Summary of Unit 10

- For an undirected graph $G = (V, E)$ with symmetric adjacency/weight matrix $A$ and degree matrix $D = \mathrm{diag}(A\mathbb{1})$, the **combinatorial Laplacian** is $L = D - A = B^T B$
- The **Laplacian quadratic form** measures smoothness across edges:

$$\boldsymbol{x}^T L \boldsymbol{x} = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}(x_i - x_j)^2 = \sum_{\{i,j\} \in E} a_{ij}(x_i - x_j)^2 \geq 0$$

- $L$ is real, symmetric and positive semi-definite with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and $L\mathbb{1} = \mathbf{0}$; the multiplicity of $\lambda = 0$ equals the number of connected components of $G$
- For any $\pm 1$ cut indicator vector $\boldsymbol{x} \in \{-1, +1\}^n$, $\mathrm{cut}(S, \bar{S}) = \frac{1}{4} \boldsymbol{x}^T L \boldsymbol{x}$
- Relaxing balanced minimum cut ($\boldsymbol{x} \perp \mathbb{1}$, $\|\boldsymbol{x}\|_2^2 = n$) to $\mathbb{R}^n$ yields the **Fiedler value** $\lambda_2$ and **Fiedler vector** $\boldsymbol{v}_2$, whose signs partition $G$ into two communities